# Integration tests and fixtures

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

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WORKBOOK · 1 OF 15

## Recall and choose your starting point

> Record your starting explanation for Integration tests and fixtures before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Testing model-dependent behaviour

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Least privilege and tool scopes

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Typed contracts and schemas

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 15

## Practise with fading help

> Complete the supported Integration tests and fixtures trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Integration test**s and **fixture**s. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

## Individual reasoning check

> Check the proposed Integration tests and fixtures explanation against the supplied facts before discussing it.

Use the worked case for **Integration test**s and **fixture**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 15

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Apply a new case independently

Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 15

## W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Integration tests and fixtures facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

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WORKBOOK · 6 OF 15

## W2 Trace the worked case

> Trace the worked case. Apply the supplied Integration tests and fixtures facts and keep your reasoning.

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

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WORKBOOK · 7 OF 15

## W3 Audit the limit

> Audit the limit. Apply the supplied Integration tests and fixtures facts and keep your reasoning.

A report omits this qualification: A mock verifies declared interactions, not real provider availability or all runtime limits. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

WORKBOOK · 8 OF 15

## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Integration tests and fixtures facts and keep your reasoning.

Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

**My reasoning and evidence**

WORKBOOK · 9 OF 15

## W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Integration tests and fixtures facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

WORKBOOK · 10 OF 15

## Project checkpoint and reflection

> Connect Integration tests and fixtures to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **integration test**s and **fixture**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **integration test**s and **fixture**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

**My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 12 OF 15**

## Pictorial case practice: Integration tests and fixtures

> Catch a wrong adapter connection. A service reads a fake catalogue fixture and answers a request for item N. Explain the labelled step before checking the changed condition.

Catch a wrong adapter connection. A service reads a fake catalogue **fixture** and answers a request for item N.

**Fixture** M="map", N="notebook". No external network is involved.

AI ACADEMY / AI-6/9

# Integration tests and fixtures

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Load fixture

The test supplies fixed records M and N.



### 2. Follow identifier

The application must request N from its adapter.



### 3. Check full reply

The returned item should be notebook with source N.

**CHANGE ONE THING**

The adapter returns M when N was requested.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Follow request N through the **fixture** and adapter. Which source and item should appear in the reply, and what precisely fails when the adapter returns M?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |

| Evidence to keep | My record |
| --- | --- |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 14 OF 15

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Catch a wrong adapter connection. A service reads a fake catalogue **fixture** and answers a request for item N.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? The returned item should be notebook with source N.

> 1. Load **fixture**

2. Follow identifier

3. Check full reply

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Check full reply The returned item should be notebook with source N. **Integration test**ing checks how components connect.

Each component can work alone while their identifier exchange is wrong.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: The adapter returns M when N was requested. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Fail the integration check: map is the wrong item.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Fail the integration check: map is the wrong item. The request now reaches **evidence** for another identifier.

**Repair a missing building block**

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the **mechanism** and terminology of testing model-dependent behaviour.

Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Review Typed contracts and schemas (ai-6/3). Goal: Explain the **mechanism** and terminology of typed contracts and schemas. Short practice: Compare A's count 4 with B's count "4".

Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?. Return to this week and explain what you changed.

**Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

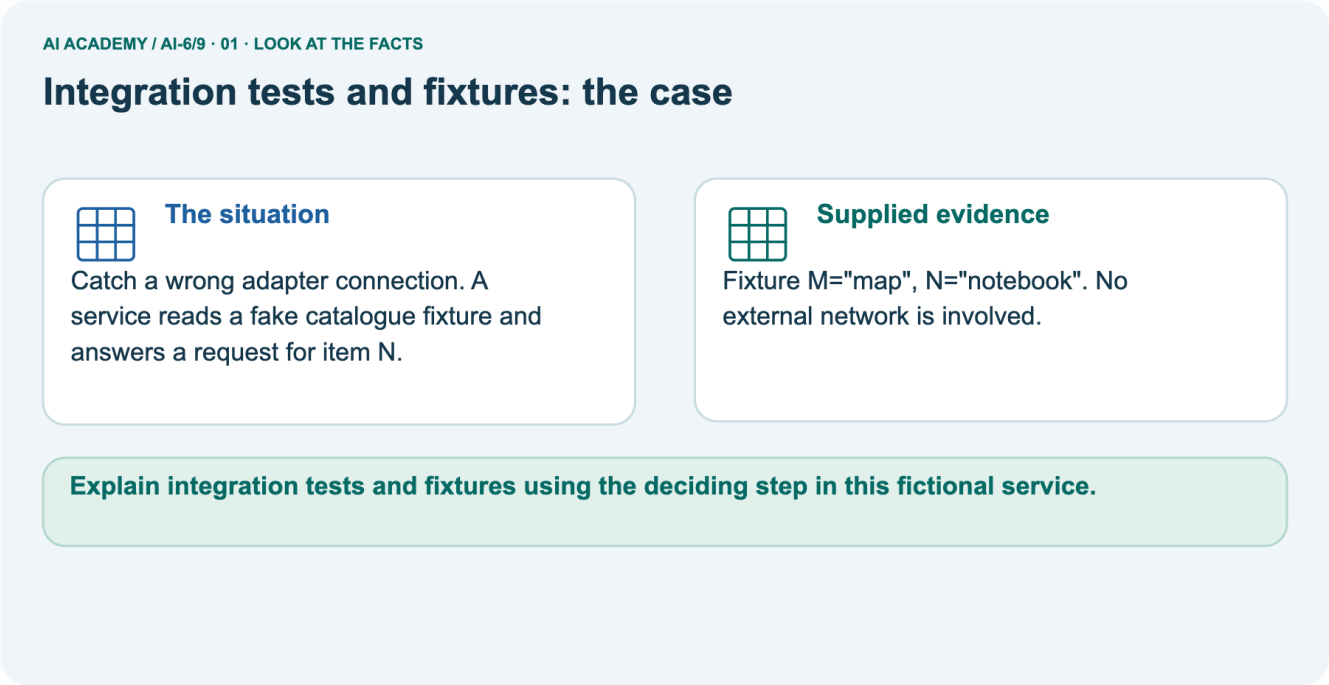
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WORKBOOK · 15 OF 15

## Visual concept journey

> Explain each Integration tests and fixtures picture from its labels, then record a relevant limit.

## 01. Integration tests and fixtures: the case



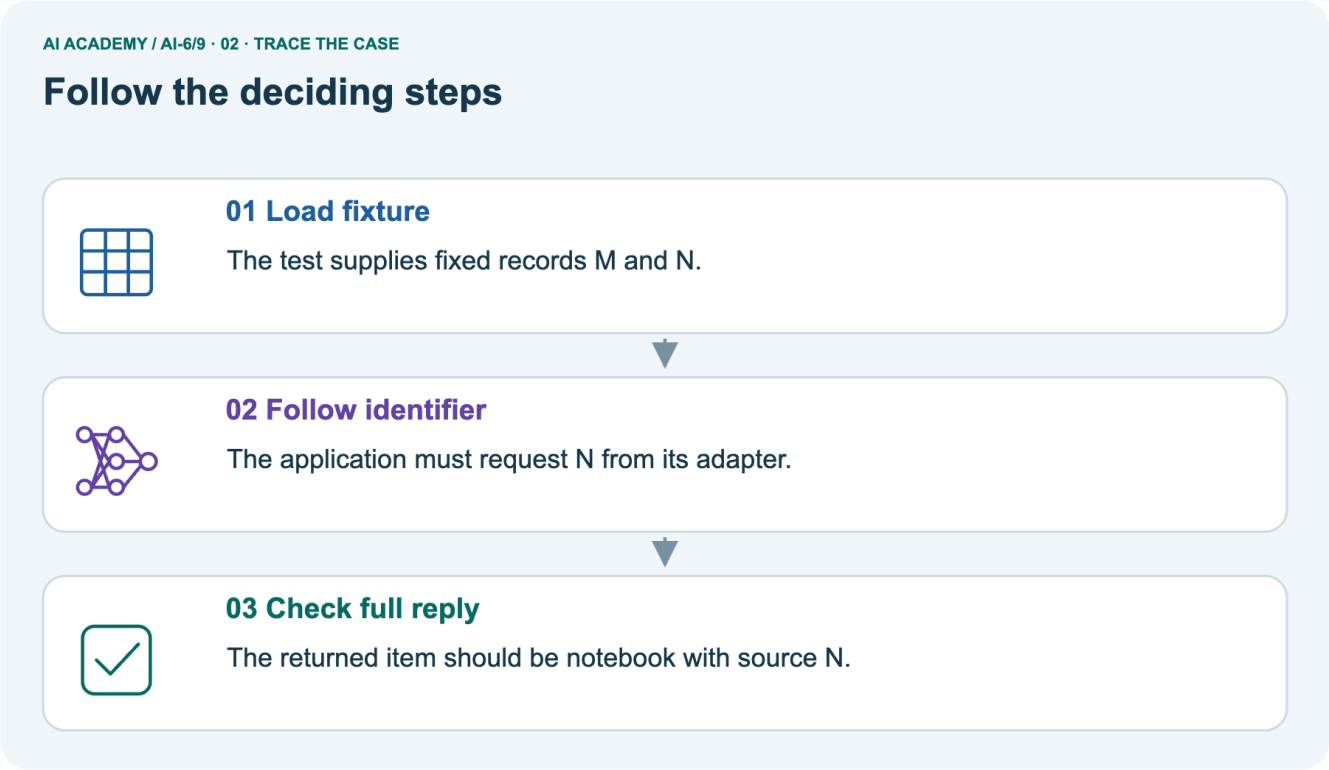

Follow the labelled picture with the matching explanation. Open the vector diagram

Catch a wrong adapter connection. A service reads a fake catalogue **fixture** and answers a request for item N. Supplied facts: **Fixture** M="map", N="notebook". No external network is involved.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



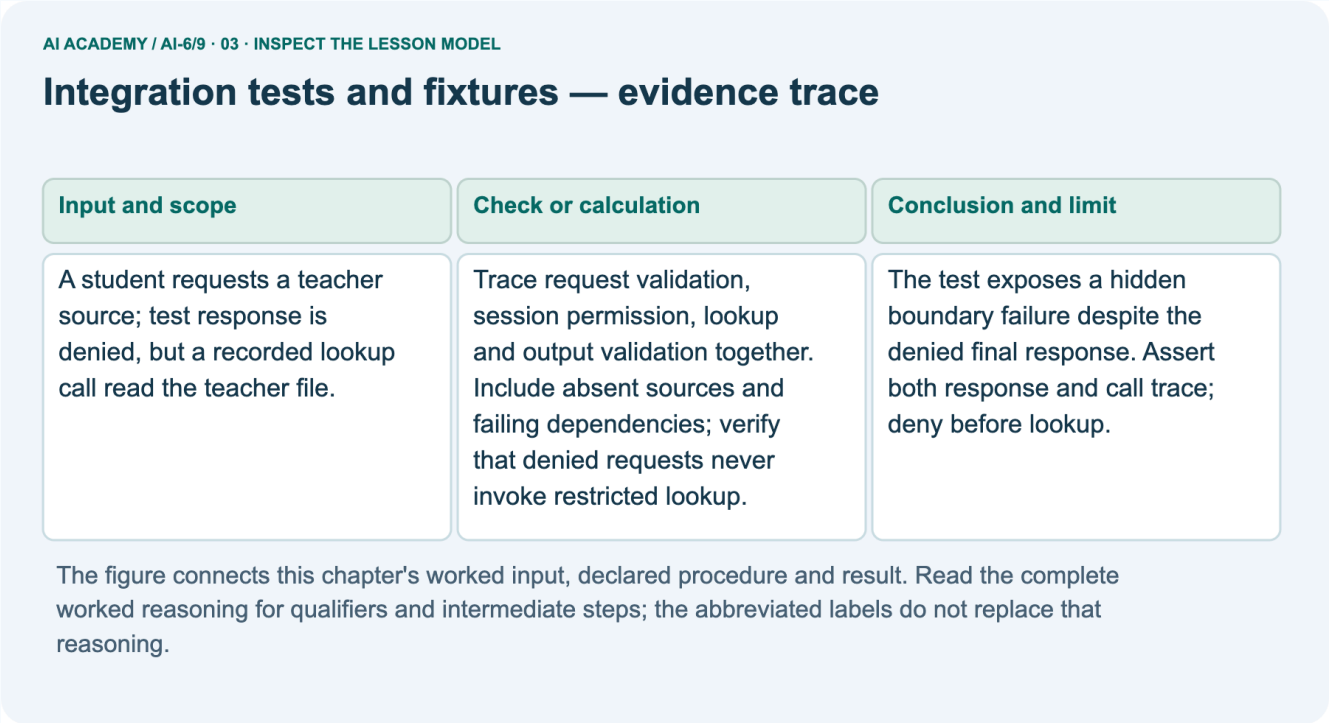

Follow the labelled picture with the matching explanation. Open the vector diagram

Load **fixture**: The test supplies fixed records M and N. → Follow identifier: The application must request N from its adapter. → Check full reply: The returned item should be notebook with source N.

> Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

### 03. Integration tests and fixtures — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

**Integration test**s and **fixture**s — **evidence** trace.

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. | Trace request validation, session permission, lookup and **output** validation together.

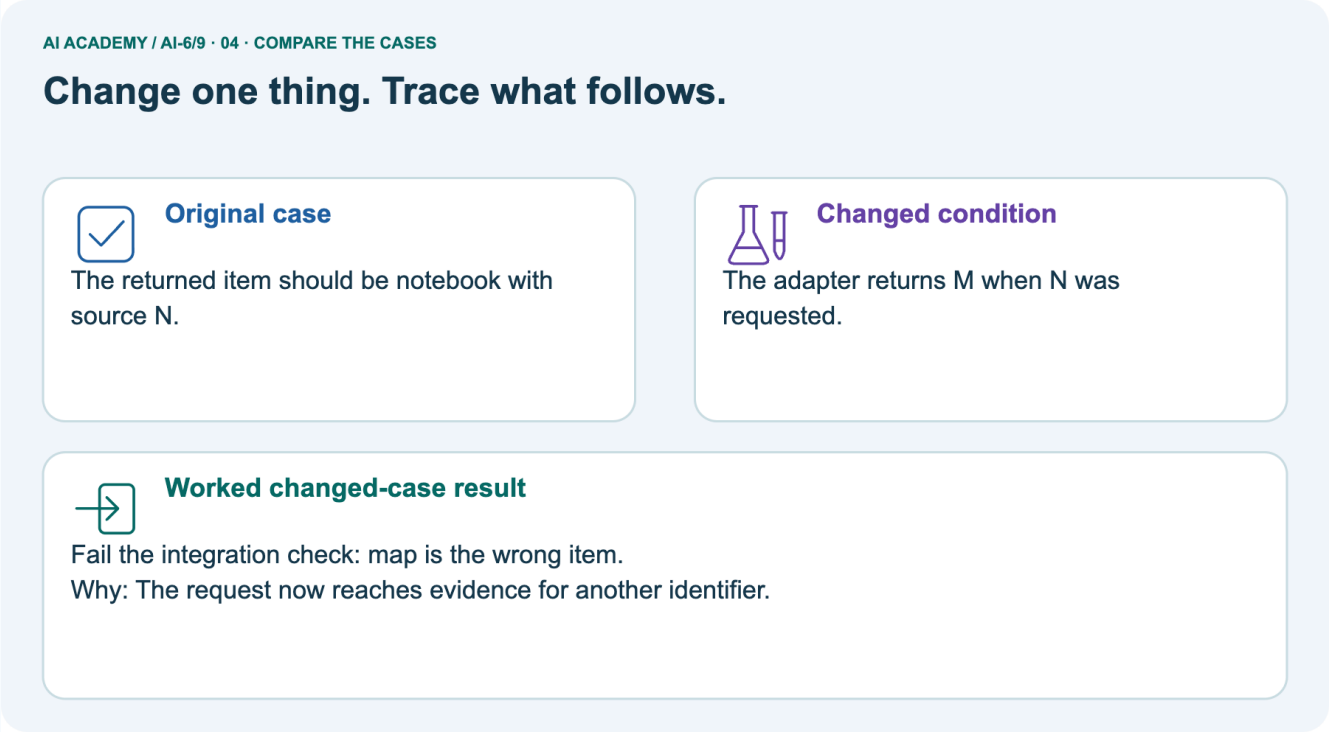
Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup. | The test exposes a hidden boundary failure despite the denied final response.

Assert both response and **call trace**; deny before lookup.

> Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

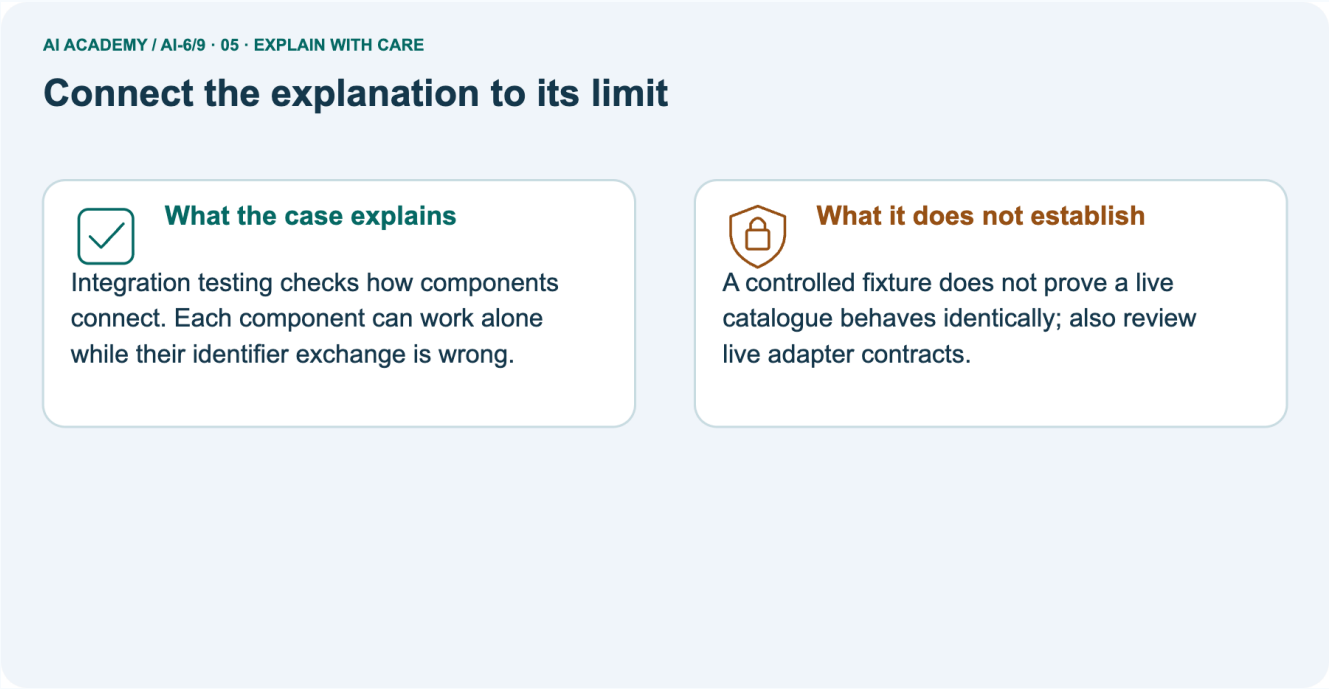
Original case: The returned item should be notebook with source N. Changed condition: The adapter returns M when N was requested. Expected result: Fail the integration check: map is the wrong item.

Reason: The request now reaches **evidence** for another identifier.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

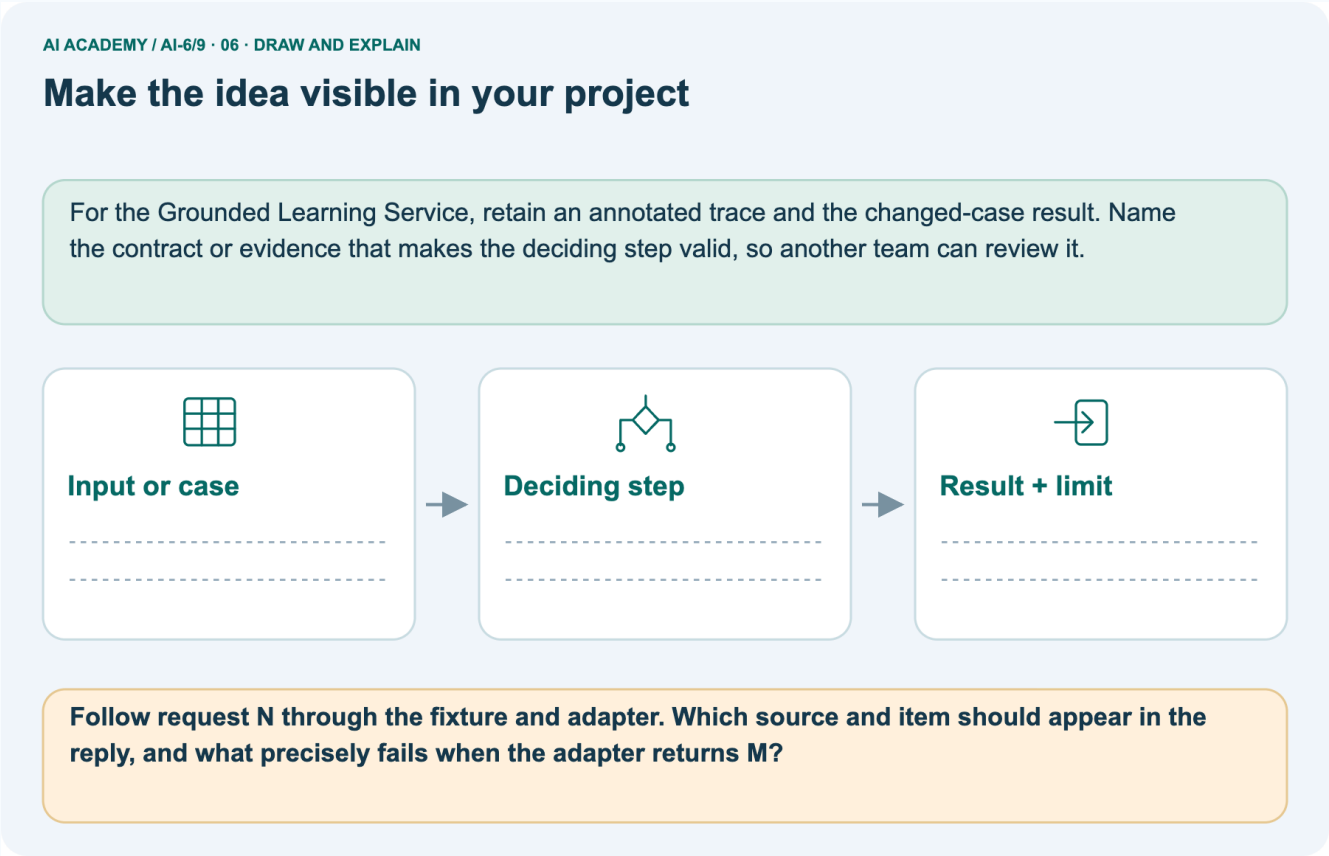
**Integration test**ing checks how components connect. Each component can work alone while their identifier exchange is wrong.

Limit: A controlled **fixture** does not prove a live catalogue behaves identically; also review live adapter contracts.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Follow request N through the **fixture** and adapter. Which source and item should appear in the reply, and what precisely fails when the adapter returns M?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.